Experiment No. 5

Feature Extraction and Image Analysis using SIFT and HOG

In [ ]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from skimage.feature import hog
from skimage import exposure
# 1. Load Image
image_path = "image.jpg.png"
image = cv2.imread(image_path)
if image is None:
    print("Image not found! Please check the image path.")
else:
    print("Image loaded successfully!")
    # 2. Convert BGR to RGB
    rgb_image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
    # Convert image to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
    # Display original and grayscale image
    plt.figure(figsize=(12, 5))
    plt.subplot(1, 2, 1)
    plt.imshow(rgb_image)
    plt.title("Original Image")
    plt.axis("off")
    plt.subplot(1, 2, 2)
    plt.imshow(gray, cmap="gray")
    plt.title("Grayscale Image")
    plt.axis("off")
    plt.show()
    # 3. SIFT Feature Extraction
    sift = cv2.SIFT_create()
    # Detect keypoints and compute descriptors
    keypoints, descriptors = sift.detectAndCompute(gray, None)
    print("\nSIFT Feature Extraction")
    print("-----------------------")
    print("Number of Keypoints:", len(keypoints))
    if descriptors is not None:
        print("Descriptor Shape:", descriptors.shape)
    # Draw SIFT keypoints
    sift_image = cv2.drawKeypoints(
        rgb_image,
        keypoints,
        None,
        flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS
    )
    # Display SIFT keypoints
    plt.figure(figsize=(8, 6))
    plt.imshow(sift_image)
    plt.title("SIFT Keypoints")
    plt.axis("off")
    plt.show()
    # 4. HOG Feature Extraction
    hog_features, hog_image = hog(
        gray,
        orientations=9,
        pixels_per_cell=(8, 8),
```

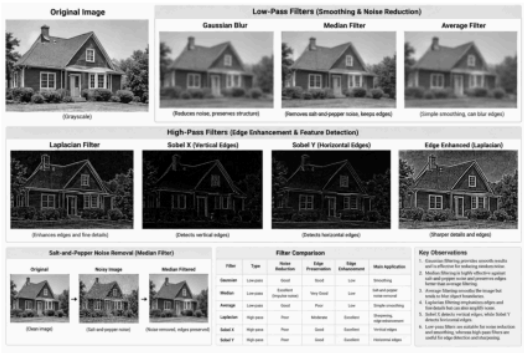
```
        cells_per_block=(2, 2),
        block_norm="L2-Hys",
        visualize=True
    )
    print("\nHOG Feature Extraction")
    print("----------------------")
    print("Number of HOG Features:", len(hog_features))
    print("HOG Feature Shape:", hog_features.shape)
    # Improve HOG visualization
    hog_image_rescaled = exposure.rescale_intensity(
        hog_image,
        in_range=(0, 10)
    )
    # Display HOG image
    plt.figure(figsize=(8, 6))
    plt.imshow(hog_image_rescaled, cmap="gray")
    plt.title("HOG Descriptor Visualization")
    plt.axis("off")
    plt.show()
    # 5. Compare SIFT and HOG
    print("\nComparison of SIFT and HOG")
    print("--------------------------")
    print("SIFT Keypoints:", len(keypoints))
    print("SIFT Descriptor Dimension:",
          descriptors.shape if descriptors is not None else "None")
    print("HOG Feature Dimension:", len(hog_features))
    print("\nSIFT:")
    print("- Detects distinctive keypoints")
    print("- Robust to scale and rotation changes")
    print("- Useful for image matching and object recognition")
    print("\nHOG:")
    print("- Represents local gradient orientations")
    print("- Captures shape and edge information")
    print("- Useful for object detection and recognition")
```

```
Image loaded successfully!
```

Original Image

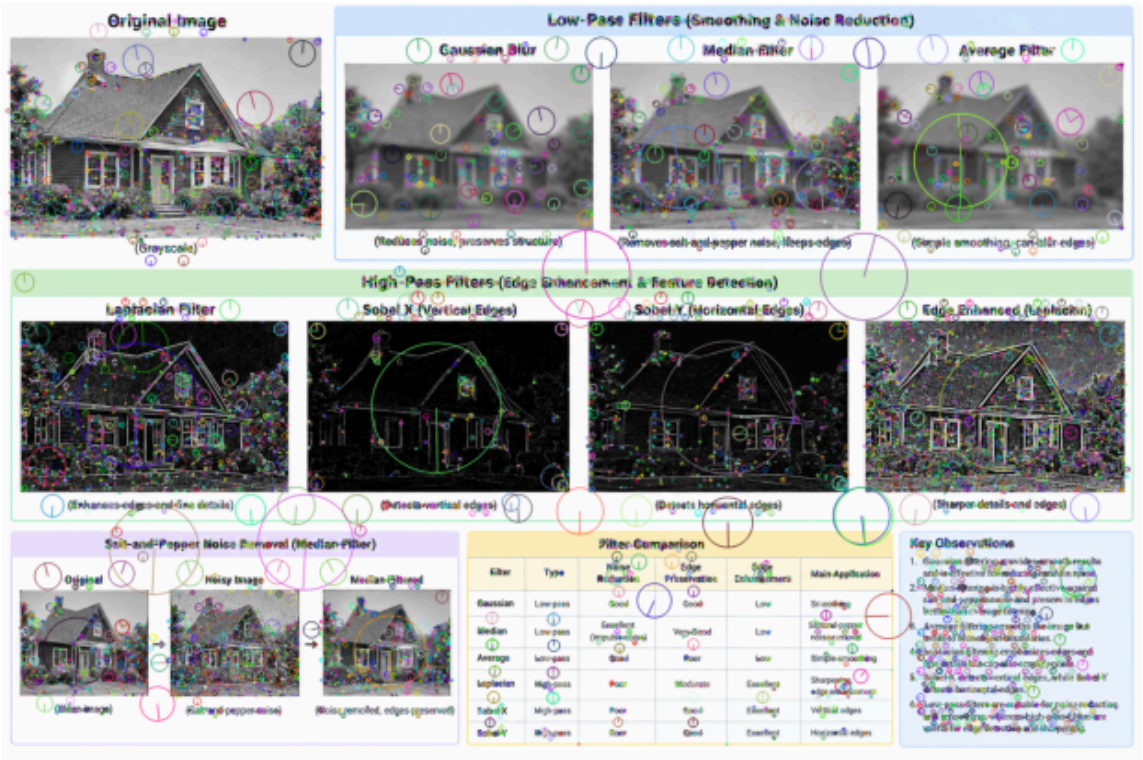


Grayscale Image




```
SIFT Feature Extraction
----------------------
Number of Keypoints: 9437
Descriptor Shape: (9437, 128)
```

SIFT Keypoints




```
HOG Feature Extraction
----------------------
Number of HOG Features: 873252
HOG Feature Shape: (873252,)
```

HOG Descriptor Visualization



```
Comparison of SIFT and HOG
--------------------------
SIFT Keypoints: 9437
SIFT Descriptor Dimension: (9437, 128)
HOG Feature Dimension: 873252

SIFT:
- Detects distinctive keypoints
- Robust to scale and rotation changes
- Useful for image matching and object recognition

HOG:
- Represents local gradient orientations
- Captures shape and edge information
- Useful for object detection and recognition
```